DASE7506 MP2

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1.Tasks and Data

This task involves selecting the correct final answer from a given mathematical problem and four candidate solutions generated by large language models. Utilizing the course-provided 1024-dimensional Qwen3-0.6B model, subtract the mean of the four candidate responses for the same question to obtain contrastive features, and train a ridge linear probe to predict whether a candidate is correct; the requirement is to select only one candidate answer per question. The official scorer parses the final numerical value of the selected candidate answer and compares it with the reference value; any correct candidate will receive a score, while the reasoning process itself is not scored. External data or pretrained weights must not be used.

This dataset comprises 4,755 training examples, 798 validation examples, and 467 test examples, each accompanied by four candidate trajectories and fixed 1024-dimensional representations generated by the Qwen3-0.6B model. The training set is utilized for model fitting, the validation set for development and debugging, while the test set is reserved exclusively for final evaluation. We deliberately maintain methodological simplicity: employing linear probes based on frozen representations, intra-example contrastive transformations, and hard-coded answer format filtering. The complete reasoning process consumes approximately 12 KiB of resources, with prediction times ranging from 0.1 to 0.2 seconds across four CPU threads.

The following were accomplished: (i) a simple and reproducible selector achieving an accuracy of 83.30% on the test set; (ii) ablation experiments isolating the effects of within-question contrast, answer explanations, and integer answer priors individually; (iii) a reliability analysis elucidating why validation accuracy performs poorly on this benchmark, identifying sources of error, and assessing the effectiveness of probe confidence in question ranking.

I focus on the third question in the Guide: Should a solution be evaluated in isolation or compared against the other three solutions to the same problem? Assume that the original trajectory representation mixes problem-level information (topic, difficulty, length) with solution-level quality; therefore, isolated scoring is noisy. Subtracting the mean of solutions to the same problem removes the common component, sharpening the comparison. Thus, I expect the contrast probe to outperform a similarly trained probe on the original embeddings, especially on problems with only one correct candidate where decision-making is most difficult.

Each question presents four fixed-order candidates, with each candidate corresponding to a 1024-dimensional vector, where the text and representations are aligned positionally. The evaluation metric is selection accuracy: the proportion of questions for which the final answer of the selected candidate, after parsing by the official parser, matches the standard answer.

Equivalent fractions and decimals are normalized, thus 0.5 and 1/2 are considered identical.

Table 1. Split composition and the number of correct candidates per group (k).

Split	Questions	k = 4	k = 3	k = 2	k = 1	Majority acc. (%)
Train	4,755	2,242	1,070	732	711	—
Validation	798	379	179	123	117	86.22
Test	467	0	171	145	151	69.16

There are significant differences in the splits. The validation set contains numerous simple questions where all four candidates are correct, whereas the test set has none of these. Each question in the test set has at least two distinct valid answers, with approximately one-third of the questions having one, two, or three correct candidates, respectively.

2. Methods

2.1 Frozen Representations

Only use embeddings provided by the course; do not invoke external models or load additional weights. Each candidate is a vector x_i in R^1024. We are unaware of the specific pooling method used by the organizers; experimentally, this vector carries a strong "correctness" signal.

2.2 Comparative Analysis of Similar Topics

The original vectors of the four candidates for the same question share the information inherent in the question itself. To eliminate this common component, we subtract the mean vector of the question: $x_i' = x_i - (1/4)\ \Sigma_j x_j$. The sum of the four contrast vectors is zero, so the scores depend solely on "the difference between the candidate and the other candidates of the same question." On the validation set, this transformation improves the probe's accuracy from 89.35% to 90.60%.

2.3 Linear Correctness Probes

For each training candidate we create a binary label $y_i = 1$ if its parsed final answer equals the reference and 0 otherwise. After standardising each feature dimension with training statistics, we fit a ridge regression: $\min_w ||Zw - (y - \bar{y})||^2 + \lambda||w||^2$, with $\lambda = 1000$. The closed-form solution is $w = (Z^TZ + \lambda I)^{-1}Z^T(y - \bar{y})$. At inference the score of candidate i is $s_i = w \cdot z_i + \bar{y}$, and the selected candidate is $\text{argmax}_i\ s_i$ within the question.

2.4 Answer format filtering

There are two kinds of candidates that can 't score regardless of reasoning. First, when parse _ answer returns None, the official scorer can 't read the final answer. Second, in all the

segmentations provided by the course, the correct final answer is a non-zero integer : in the training set, about 1,200 non-integer answers and 98 candidates with zero answers are not correct, while 12 negative integers are correct. Therefore, I only use the ' answer is a non-zero integer ' candidate training, and the remaining candidates are shielded when reasoning, and the negative integers are still retained.

2.5 Training and reasoning.

The probe is fitted only once on the training set, and the validation set is only used to select the model. Inference assets save w, feature mean and standard deviation, intercept and feature mode, a total of about 12KiB.One prediction is that each bank performs a matrix multiplication and then takes argmax in each question ; the 467 test questions take about 0.1-0.2 seconds on a four-core CPU.

3. Experiment

3.1 Settings

All numbers are obtained by the official parser and scorer. We fit on the training set, select hyperparameters on the validation set, and report only once on the test set after freezing the method. The baseline includes : always select the first, vote for the majority of the analytical answer (flat votes take the minimum subscript), select the longest or shortest trajectory. As ablation, probes on the original embedding, and probes with only ' unparsable mask ' without integer filtering are also reported.

Table 2. Main results. Validation is used for development; test is scored once for the frozen method. Dashes indicate methods not evaluated on test.

Method	Validation (%)	Test (%)
Always choose first candidate	78.20	49.68
Majority vote	86.22	69.16
Longest trace	77.94	54.82
Shortest trace	69.30	38.76
Raw probe (λ = 1000)	89.35	—
Contrast probe	90.60	—
+ unparseable mask (v1)	91.35	82.01
+ integer-answer filter (v3)	93.36	83.30

3.2 Results and ablation

The original probe has exceeded the voting baseline on the validation set, with a 1.25 % increase in contrast features, a 0.75 % increase in screening non-parsable candidates, and a 2.0 % increase in limiting training and reasoning to non-zero integer answers. On the test set, the latter two versions are 82.01 and 83.30, respectively, DASE7506MP2 · Choose reliable LLM

reasoning · 3 integer filtering value 1.29 %. The relative order of each version is consistent in two segmentations, but the absolute difference between the validation set and the test set is very large, which supports my hypothesis.

4. Error analysis

4.1 The validation set is not a small test set.

There are 379 questions in the validation set, of which four candidates are all pairs ; therefore, a method that seems to have an accuracy of 93 % on the validation set may have an accuracy of only 83 % on the test set, and a method that performs better on the validation set may perform worse on the test set.

4.2 source of error

There are 78 errors in the freezing model on the test set. The error is highly concentrated on the question with only one correct candidate, and 151 such questions contribute 49 errors. Most of the votes on these questions are almost invalid, because the three wrong answers are often consistent with each other ; when there are two correct candidates, the probe is close to the vote ; when there are three correct candidates, the votes are all right, and the probe is slightly reduced.

Table 3. Error analysis on the test split by group composition.

Correct candidates	Groups	Errors	Share of errors (%)	Probe acc. (%)	Majority acc. (%)
1	151	49	62.8	67.5	20.5
2	145	21	26.9	85.5	83.4
3	171	8	10.3	95.3	100.0
All	467	78	100.0	83.3	69.2

39. 7 % of the test errors selected answers that appeared at least twice in the four candidates. The remaining errors selected a small number of answers.

4.3 The model relies on

The length is not a reliable clue : the longest trajectory is selected with a score of 77.94 on the validation set and only 54.82 on the test set ; the shortest trajectories are selected with scores of 69.30 and 38.76 on the validation set and the test set, respectively, and the probe is obviously superior to these surface baseline methods, and the contrast transformation enables it to compare candidate answers rather than judge them in isolation. Probes may use confident wording or format related to correctness in training. Integer answer filtering is a format priori superimposed on the verifier and is not evidence of reasoning ability.

5. Discussion and limitations

It is mainly about the diagnostic value : it shows the extent to which the linear model can achieve in the case of frozen hidden states, and the source of residual errors. There are three main limitations. Three limitations are worth emphasizing. First, the integer answer prior is only applicable to GSM8K-style data ; in the data set containing scores or zero answers, this method will discard the correct answers.Secondly, the group composition of the verification set and the test set is different, so the verification accuracy cannot be used as an estimation of the test accuracy. Our development decisions depend on relative comparisons and more difficult verification subsets. Third, the test set only evaluates the freezing method once, but the test reference is public, and there is a risk of overfitting by repeatedly viewing test errors .

The submission consists of submission.py (prediction interface), train_probe.py (training), runs/model_bundle/probe.npz (12 KiB checkpoint), runs/test-score.json (official score) and this report. All arelinked from the public repository; the checkpoint reproduces 83.2976 on test without retraining.

AI assistance was used for code implementation, analysis scripts.

Appendix A. Official evaluation logs

The screenshots below are raw terminal outputs of the official scorer.

(a) Majority vote, validation: 86.2155% (688/798).

```
(.venv) C:\Users\莫默沫墨\Desktop\DASE7506-MP2-main\code>notepad submission.py

(.venv) C:\Users\莫默沫墨\Desktop\DASE7506-MP2-main\code>python predict.py --method sub
mission.py --split validation --output runs\validation-predictions.json
{"protocol": "dase7506-mp2-open-hard-verifier-v5", "split": "validation", "prediction_s
econds": 0.11184560000401689}

(.venv) C:\Users\莫默沫墨\Desktop\DASE7506-MP2-main\code>python score_predictions.py --
split validation --predictions runs\validation-predictions.json --output runs\validatio
n-score.json
{
  "protocol": "dase7506-mp2-open-hard-verifier-v5",
  "split": "validation",
  "correct": 688,
  "n": 798,
  "accuracy": 86.2155388471178,
  "scoring_seconds": 0.04034529999626102
}
```

(b) Final method, validation: 93.3584% (745/798).

```
(.venv) C:\Users\莫默沫墨\Desktop\DASE7506-MP2-main\code>python score_predictions.py --
split validation --predictions runs\validation-predictions.json --output runs\validatio
n-score.json
{
  "protocol": "dase7506-mp2-open-hard-verifier-v5",
  "split": "validation",
  "correct": 745,
  "n": 798,
  "accuracy": 93.35839598997494,
  "scoring_seconds": 0.0777912000194192
}
```

(c) Final method, test: 83.2976% (389/467).

```
(.venv) C:\Users\莫默沫墨\Desktop\DASE7506-MP2-main\code>python predict.py --method sub
mission.py --checkpoint runs\model_bundle --split test --output runs\test-predictions.j
son
{"protocol": "dase7506-mp2-open-hard-verifier-v5", "split": "test", "prediction_seconds
": 0.07671209998079576}

(.venv) C:\Users\莫默沫墨\Desktop\DASE7506-MP2-main\code>python score_predictions.py --
split test --predictions runs\test-predictions.json --output runs\test-score.json
{
  "protocol": "dase7506-mp2-open-hard-verifier-v5",
  "split": "test",
  "correct": 389,
  "n": 467,
  "accuracy": 83.29764453961457,
  "scoring_seconds": 0.020868799998424947
}
```